Task 4: Cloud Platform Exploration

Platforms covered: AWS, Microsoft Azure, Google Cloud Platform (GCP), Cloudflare

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1. Introduction

Cloud computing lets teams rent computing resources (servers, storage, databases, networking, AI tooling) over the internet and pay only for what they use, instead of buying and running their own hardware. This document explores four major platforms, explains what their most important services do and how they are typically used, and lists the official documentation and learning resources for each.

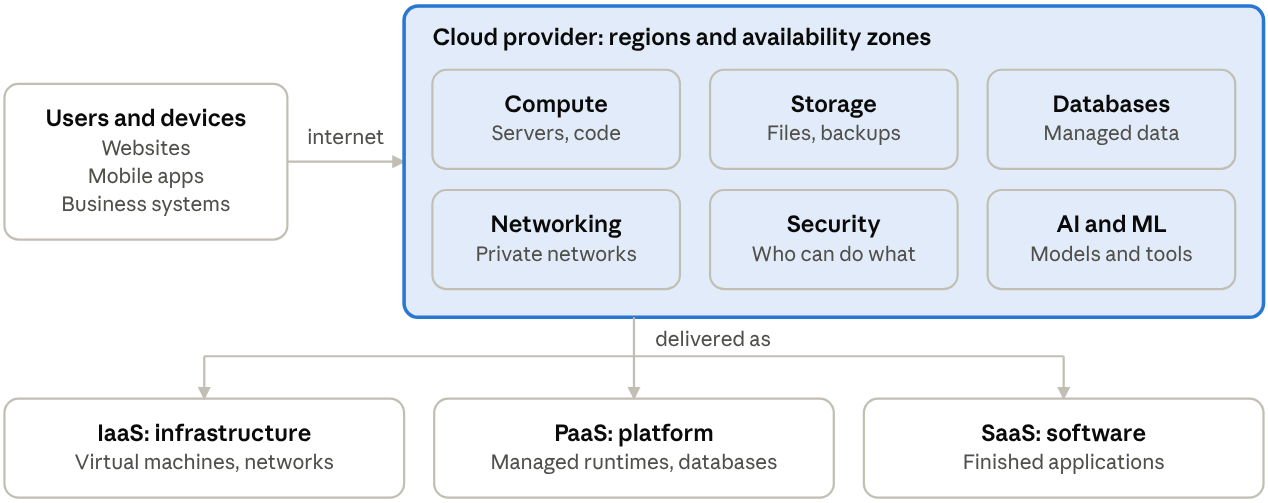
Three of them (AWS, Azure, GCP) are **hyperscale general-purpose clouds** offering hundreds of services. Cloudflare is different: it began as a security and content-delivery network and has grown into an **edge cloud** that runs code and stores data close to users around the world.

Key cloud concepts used throughout

- **IaaS, PaaS, SaaS:** Infrastructure (virtual machines, networks), Platform (managed runtimes and databases), Software (finished applications). Most services below fall into IaaS or PaaS.
- **Region and Availability Zone:** A region is a geographic area; an availability zone is an isolated data center (or group of them) inside a region. Deploying across zones gives resilience.
- **Shared responsibility model:** The provider secures the underlying cloud; the customer secures what they put in it (configuration, data, access).
- **Serverless:** You deploy code or queries and the provider handles servers, scaling, and patching. Billing is per request or per execution time.
- **Pay-as-you-go pricing:** Billed by usage, with discounts for committed use and free tiers for learning.

Cloud computing at a glance

Cloud computing delivers shared resources over the internet, on demand

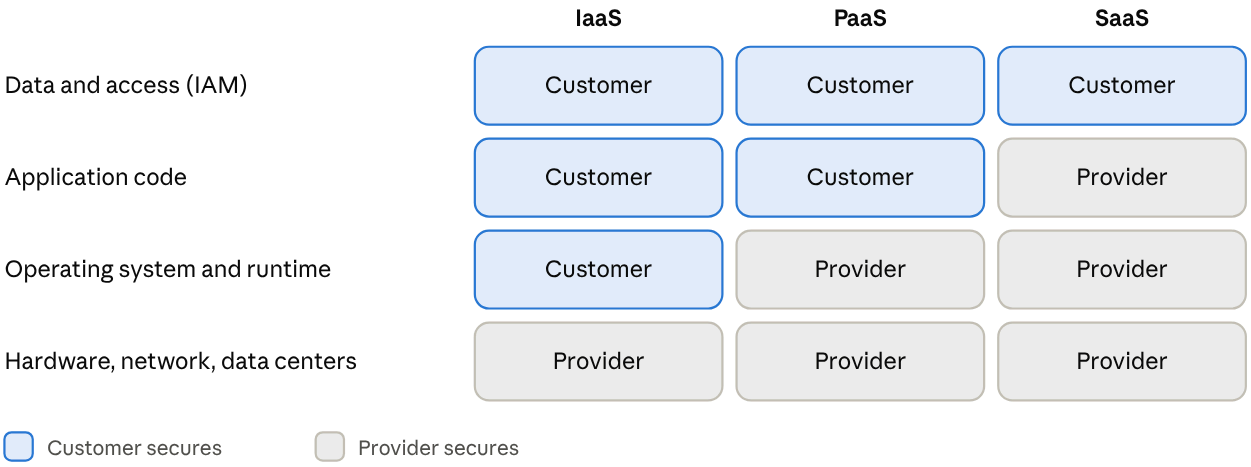


cloud computing overview · 6 service areas, 3 service models

Users reach the provider's resources over the internet, and the service model (IaaS, PaaS or SaaS) sets how much they manage themselves.

Shared responsibility model

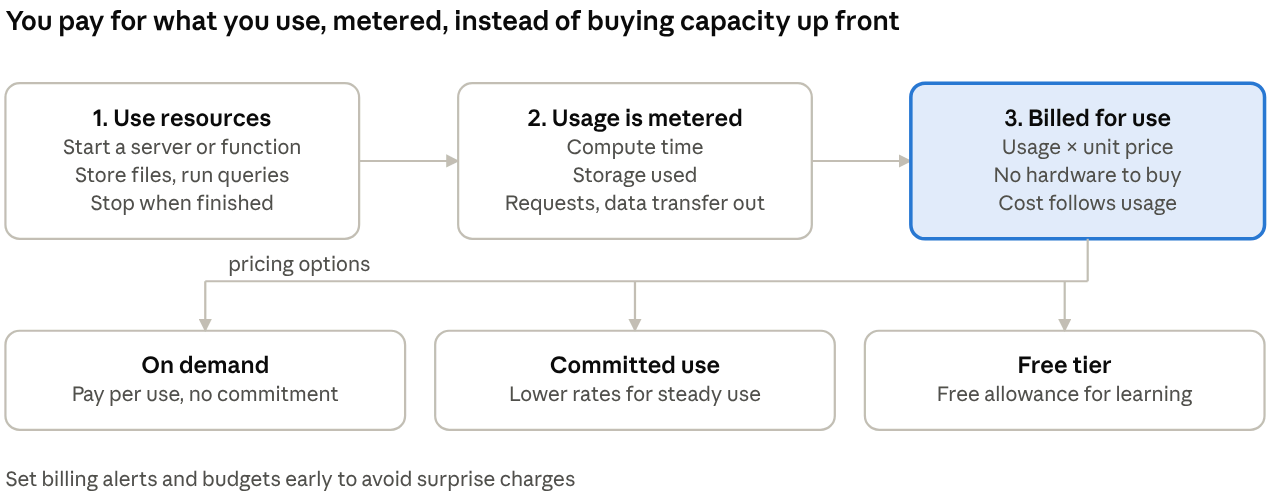
The provider secures the cloud; the customer secures what runs in it



shared responsibility model · 4 layers by 3 service models

The provider always secures the hardware and data centers; the customer always secures data and access. The split in between moves toward the provider as the service model goes from IaaS to SaaS.

Pay-as-you-go pricing



pay-as-you-go pricing · 3 steps, 3 pricing options

Charges follow metered usage; committed-use discounts and free tiers change the rate or the first units, not the model.

2. Amazon Web Services (AWS)

Overview

AWS launched in 2006 and is the largest and most mature cloud provider, with the broadest service catalog and a very large global infrastructure of regions and availability zones. It is a common default for startups and enterprises alike.

Important services and how they are used

Compute

- **EC2 (Elastic Compute Cloud):** Virtual machines in many sizes. Used for web servers, backend services, batch jobs, or anything needing full control of the operating system.

- **Lambda:** Serverless functions triggered by events (an HTTP request, a file upload, a queue message). Used for APIs, automation, and event-driven pipelines.
- **ECS, EKS, Fargate:** Container services. ECS is AWS's own orchestrator, EKS is managed Kubernetes, and Fargate runs containers without managing servers.
- **Elastic Beanstalk and App Runner:** Simpler platforms for deploying web apps without managing infrastructure details.

Storage

- **S3 (Simple Storage Service):** Object storage for files, backups, static website assets, data lakes, and logs. One of the most widely used AWS services.
- **EBS:** Block storage disks attached to EC2 instances.
- **EFS:** Managed shared file system for multiple instances.
- **S3 Glacier:** Low-cost archival storage for rarely accessed data.

Databases

- **RDS and Aurora:** Managed relational databases (PostgreSQL, MySQL, and others). Aurora is AWS's high-performance cloud-native engine.
- **DynamoDB:** Fully managed NoSQL key-value and document database with very low latency at scale.
- **ElastiCache:** Managed Redis and Memcached for caching.
- **Redshift:** Data warehouse for analytics.

Networking and delivery

- **VPC (Virtual Private Cloud):** Your private network in AWS: subnets, route tables, gateways, and security groups.
- **Route 53:** DNS and domain registration with health checks and routing policies.
- **CloudFront:** Content delivery network (CDN) that caches content near users.
- **Elastic Load Balancing and API Gateway:** Distribute traffic and expose managed APIs.

Security and identity

- **IAM (Identity and Access Management):** Controls who (users, roles, services) can do what. The foundation of AWS security.
- **KMS and Secrets Manager:** Encryption key management and secret storage.
- **WAF, Shield, GuardDuty:** Web firewall, DDoS protection, and threat detection.

Integration, monitoring, and DevOps

- **SQS, SNS, EventBridge:** Queues, notifications, and event routing to decouple applications.
- **CloudWatch and CloudTrail:** Metrics, logs, and alarms; and an audit log of API activity.
- **CloudFormation and CDK:** Infrastructure as code.
- **CodePipeline, CodeBuild, CodeDeploy:** CI/CD tooling.

AI and machine learning

- **SageMaker:** Build, train, and deploy ML models.
- **Bedrock:** Managed access to foundation models from multiple providers for building generative AI applications.

Typical use case

A web application might run on containers (ECS or EKS) behind a load balancer inside a VPC, store data in Aurora, serve images from S3 through CloudFront, and be monitored with CloudWatch, with all access controlled through IAM.

Documentation and learning resources

- Official documentation: docs.aws.amazon.com
- AWS Skill Builder (free and paid training, labs, exam prep): skillbuilder.aws
- AWS Well-Architected Framework (best practices across six pillars)
- AWS Free Tier (free offers for new accounts, including credits for new customers)
- Certifications: Cloud Practitioner (entry level), Solutions Architect Associate, Developer Associate, and higher tiers
- AWS Architecture Center and AWS Workshops for hands-on tutorials

3. Microsoft Azure

Overview

Azure is Microsoft's cloud platform and the second-largest by market share. Its biggest strength is integration with the Microsoft ecosystem (Windows Server, Active Directory, SQL Server, .NET, Microsoft 365, GitHub), which makes it especially popular in enterprises and for hybrid setups that combine on-premises servers with the cloud.

Important services and how they are used

Compute

- **Virtual Machines:** Windows and Linux VMs for general workloads and lift-and-shift migrations.
- **Azure App Service:** Managed hosting for web apps and APIs in .NET, Node.js, Python, Java, and more, without managing servers.
- **Azure Functions:** Serverless, event-driven code.
- **AKS (Azure Kubernetes Service)** and **Azure Container Apps:** Managed Kubernetes and a simpler serverless container platform.
- **Virtual Machine Scale Sets:** Automatically scale groups of VMs.

Storage

- **Blob Storage:** Object storage for unstructured data such as images, backups, and data lake files.
- **Azure Files:** Managed file shares.
- **Managed Disks:** Block storage for VMs.

Databases

- **Azure SQL Database and SQL Managed Instance:** Managed SQL Server.
- **Azure Database for PostgreSQL and MySQL:** Managed open-source relational databases.
- **Cosmos DB:** Globally distributed multi-model NoSQL database.
- **Azure Cache for Redis:** Managed caching.
- **Synapse Analytics and Microsoft Fabric:** Analytics and data platform services.

Networking

- **Virtual Network (VNet):** Private networks with subnets and security rules.
- **Azure Load Balancer, Application Gateway, and Front Door:** Regional and global traffic distribution, with Front Door also providing CDN and WAF capabilities.
- **Azure DNS, ExpressRoute, VPN Gateway:** DNS hosting and private or encrypted connectivity to on-premises networks.

Identity and security

- **Microsoft Entra ID (formerly Azure Active Directory):** Cloud identity, single sign-on, and multi-factor authentication. A core reason enterprises choose Azure.

- **Key Vault:** Secrets, keys, and certificates.
- **Microsoft Defender for Cloud and Azure Policy:** Security posture management and governance rules.

DevOps and management

- **Azure DevOps and GitHub Actions:** Repositories, pipelines, and boards for CI/CD.
- **Azure Monitor, Log Analytics, Application Insights:** Monitoring, logging, and application performance tracking.
- **ARM templates and Bicep (and Terraform):** Infrastructure as code.
- **Azure Resource Manager:** The management layer where resources are organized into resource groups, subscriptions, and management groups.

AI and machine learning

- **Azure Machine Learning:** Train, manage, and deploy ML models.
- **Azure AI Foundry and Azure OpenAI:** Build generative AI applications with models and tools, within Azure's security and compliance controls.
- **Azure AI Services:** Prebuilt APIs for vision, speech, language, and document processing.

Typical use case

An enterprise might extend its on-premises Active Directory to Entra ID, host an internal .NET application on App Service connected to Azure SQL, store documents in Blob Storage, and monitor everything with Azure Monitor.

Documentation and learning resources

- Official documentation: learn.microsoft.com/azure
- Microsoft Learn (free learning paths, modules, and sandboxes)
- Azure Architecture Center and the Azure Well-Architected Framework
- Azure Free Account (free tier services plus starter credit for new users)
- Certifications: AZ-900 Azure Fundamentals (entry level), AZ-104 Administrator, AZ-204 Developer, AZ-305 Solutions Architect
- Azure portal, Azure CLI, and Azure PowerShell for hands-on practice

4. Google Cloud Platform (GCP)

Overview

GCP runs on the same infrastructure Google uses for Search, YouTube, and Gmail. It is known for strengths in data analytics, machine learning, Kubernetes (which Google originally created), and a developer-friendly, global private network. It is often chosen by data-heavy and AI-focused teams.

Important services and how they are used

Compute

- **Compute Engine:** Virtual machines, including options for GPUs and Google's TPUs.
- **Cloud Run:** Run containers serverlessly with automatic scaling, including down to zero. Very popular for APIs and web services.
- **Cloud Functions:** Event-driven serverless functions.
- **Google Kubernetes Engine (GKE):** Widely regarded as a leading managed Kubernetes service.
- **App Engine:** Older fully managed platform for web apps.

Storage

- **Cloud Storage:** Object storage with storage classes from frequently accessed to archival.
- **Persistent Disk and Filestore:** Block storage and managed file storage.

Databases and analytics

- **Cloud SQL and AlloyDB:** Managed MySQL, PostgreSQL, and SQL Server; AlloyDB is a high-performance PostgreSQL-compatible option.
- **Cloud Spanner:** Globally distributed relational database with strong consistency.
- **Firestore and Bigtable:** Document NoSQL database for apps, and wide-column NoSQL for very large, low-latency workloads.
- **BigQuery:** Serverless data warehouse for running SQL analytics over huge datasets. One of GCP's flagship services.
- **Pub/Sub, Dataflow, Dataproc:** Messaging, stream and batch data processing, and managed Spark and Hadoop.

Networking

- **VPC:** Global virtual networks (a VPC spans regions, which differs from AWS and Azure).
- **Cloud Load Balancing, Cloud CDN, Cloud DNS:** Global load balancing, content delivery, and DNS.
- **Cloud Armor:** DDoS protection and web application firewall.

Identity and security

- **Cloud IAM:** Permissions for users, groups, and service accounts, applied across the resource hierarchy (organization, folders, projects).
- **Cloud KMS and Secret Manager:** Encryption keys and secrets.
- **Security Command Center:** Security and risk visibility.

DevOps and operations

- **Cloud Build and Artifact Registry:** CI/CD builds and container or package storage.
- **Cloud Monitoring and Cloud Logging:** Observability tools.
- **Terraform and Infrastructure Manager:** Infrastructure as code.

AI and machine learning

- **Vertex AI:** Unified platform to build, tune, and deploy ML and generative AI models, including access to Google's Gemini models.
- **Prebuilt AI APIs:** Vision, speech, translation, and document processing.

Typical use case

A product team might deploy an API on Cloud Run, store user data in Cloud SQL, stream events through Pub/Sub into BigQuery for analytics, and use Vertex AI to add recommendations or generative features.

Documentation and learning resources

- Official documentation: cloud.google.com/docs
- Google Cloud Skills Boost (hands-on labs and learning paths)
- Google Cloud Architecture Center and Architecture Framework
- Google Cloud Free Program (free trial credit for new customers plus an Always Free tier for selected products)
- Certifications: Cloud Digital Leader (entry level), Associate Cloud Engineer, Professional Cloud Architect, and others

- Google Cloud Console, Cloud Shell, and the gcloud CLI for practice

5. Cloudflare

Overview

Cloudflare operates one of the world's largest networks, with data centers in hundreds of cities. Traditionally it sits **in front of** a website to make it faster and safer: it handles DNS, caches content, and blocks attacks. It has since expanded into a developer platform where you can run code, store data, and use AI directly at the edge, close to the user, rather than in a few central regions. It is not a full replacement for AWS, Azure, or GCP, but is often used alongside them or for edge-first applications.

Important services and how they are used

Application performance and delivery

- **CDN:** Caches static content at edge locations so pages load faster and origin servers face less traffic.
- **DNS:** Fast authoritative DNS, usually the first thing a site owner moves to Cloudflare. Also offers the public 1.1.1.1 resolver.
- **Load Balancing, Argo Smart Routing, and Cache Rules:** Traffic steering, optimized routing, and fine control over caching.
- **Images and Stream:** Image optimization and transformation, and video storage and streaming.

Security

- **DDoS protection:** Always-on mitigation of large volumetric attacks, included even on the free plan.
- **WAF (Web Application Firewall):** Blocks common web attacks such as SQL injection and cross-site scripting.
- **Bot Management and Rate Limiting:** Detect scrapers and abusive bots, and cap request rates.
- **SSL/TLS:** Free certificates and HTTPS enforcement.
- **Turnstile:** A privacy-friendly CAPTCHA alternative.

Zero Trust and network services (Cloudflare One)

- **Access:** Protects internal apps by verifying identity before granting access, replacing traditional VPNs.
- **Gateway:** Secure DNS and web filtering for users and devices.
- **Tunnel (cloudflared):** Exposes a local or private service to the internet securely without opening inbound ports.
- **WARP and Magic WAN:** Device-level secure connectivity and wide-area networking.

Developer platform

- **Workers:** Serverless functions that run at the edge on V8 isolates, giving very fast cold starts. Used for APIs, redirects, A/B testing, and full applications.
- **Pages:** Hosting for static sites and full-stack frameworks with Git-based deployments.
- **R2:** S3-compatible object storage with no egress (data transfer out) fees, which is a major cost advantage.
- **KV:** Globally distributed key-value store for read-heavy data such as configuration.
- **D1:** Serverless SQL database built on SQLite.
- **Durable Objects:** Stateful objects for coordination, such as chat rooms, counters, and real-time collaboration.
- **Queues:** Message queues for background work between Workers.
- **Hyperdrive:** Speeds up connections to existing regional databases such as PostgreSQL.
- **Workers AI and AI Gateway:** Run open AI models on Cloudflare's network, and manage, cache, and observe calls to AI providers.
- **Vectorize:** Vector database for semantic search and retrieval.

Typical use case

A site owner points their domain to Cloudflare to get DNS, HTTPS, CDN caching, and WAF protection with no code changes. A developer might go further by deploying a Worker for an API, storing files in R2, saving data in D1, and securing an admin panel with Access.

Documentation and learning resources

- Official documentation: developers.cloudflare.com
- Cloudflare Learning Center (learning.cloudflare.com): plain-language explainers on DNS, CDN, DDoS, TLS, and networking concepts
- Cloudflare Workers tutorials, templates, and the Wrangler CLI

- Cloudflare's generous free plan and Workers free tier for experimentation
- Cloudflare blog for architecture deep dives and new feature announcements
- Cloudflare Developer Discord and community forum

6. Side-by-Side Comparison

Area	AWS	Azure	GCP	Cloudflare
Virtual machines	EC2	Virtual Machines	Compute Engine	Not offered (edge compute only)
Serverless functions	Lambda	Azure Functions	Cloud Functions / Cloud Run	Workers
Managed Kubernetes	EKS	AKS	GKE	Not offered
Object storage	S3	Blob Storage	Cloud Storage	R2
Relational database	RDS / Aurora	Azure SQL / PostgreSQL	Cloud SQL / AlloyDB / Spanner	D1 (SQLite-based)
NoSQL database	DynamoDB	Cosmos DB	Firestore / Bigtable	KV / Durable Objects
Data warehouse	Redshift	Synapse / Fabric	BigQuery	Not offered
CDN	CloudFront	Front Door / Azure CDN	Cloud CDN	Cloudflare CDN (core product)
DNS	Route 53	Azure DNS	Cloud DNS	Cloudflare DNS
Identity	IAM	Entra ID	Cloud IAM	Access (Zero Trust)
Web firewall / DDoS	WAF, Shield	Front Door WAF, DDoS Protection	Cloud Armor	WAF, DDoS (core product)

Area	AWS	Azure	GCP	Cloudflare
Monitoring	CloudWatch	Azure Monitor	Cloud Monitoring	Analytics, Logs, Workers Observability
Infrastructure as code	CloudFormation, CDK	Bicep, ARM	Terraform, Infrastructure Manager	Terraform provider, Wrangler
AI / ML	SageMaker, Bedrock	Azure ML, AI Foundry	Vertex AI	Workers AI, AI Gateway

Strengths at a glance

- **AWS:** Broadest service range, largest ecosystem and community, very mature.
- **Azure:** Best fit for Microsoft-centric organizations, strong hybrid cloud and enterprise identity.
- **GCP:** Excellent for data analytics (BigQuery), Kubernetes, and AI/ML.
- **Cloudflare:** Best-in-class edge performance and security, simple developer experience, and no egress fees on R2.

Choosing between them

The right choice depends on the workload, existing skills, budget, and compliance needs. Many organizations use more than one (multi-cloud). A common pattern is running core workloads on AWS, Azure, or GCP while using Cloudflare in front for DNS, CDN, and security.

7. Common Learning Path

1. Learn the fundamentals: regions, IAM, networking, and the shared responsibility model.
2. Create a free account on each platform and complete a beginner lab (for example, host a static site).
3. Deploy the same simple app (a small API plus a database) on each platform and compare the experience.
4. Learn one infrastructure-as-code tool, such as Terraform, to work across all four.

5. Study for one entry-level certification: AWS Cloud Practitioner, AZ-900, or Cloud Digital Leader.
6. Set billing alerts and budgets early, and delete resources after practicing to avoid unexpected charges.

8. Key Takeaways

- All three hyperscalers cover the same core building blocks (compute, storage, databases, networking, identity, AI), but with different names and design philosophies.
- Learning one makes the others much easier, since the concepts map closely.
- Cloudflare complements them by moving security, caching, and compute to the edge, and its developer platform is growing quickly.
- Official documentation and the free hands-on learning portals are the best way to go deeper.

9. References

- AWS Documentation: docs.aws.amazon.com
- AWS Skill Builder: skillbuilder.aws
- Microsoft Azure Documentation: learn.microsoft.com/azure
- Google Cloud Documentation: cloud.google.com/docs
- Google Cloud Skills Boost: cloudskillsboost.google
- Cloudflare Developer Docs: developers.cloudflare.com
- Cloudflare Learning Center: cloudflare.com/learning

Note: Service names, free-tier terms, and pricing change often, so confirm current details on each provider's official site before relying on them.